

Prompt Engineering in Large Language Models (LLMs)

What Is a Prompt?

A **prompt** is the input text given to a large language model (LLM), such as GPT, to initiate a response. It can be as simple as a single question or as complex as a structured input with multiple examples and instructions.

```
Translate to French:  
I love apples → J'aime les pommes  
I love music →
```

What Is Prompt Engineering?

Prompt engineering is the process of designing and structuring prompts to guide a language model toward producing useful, accurate, or creative outputs. Rather than changing the model's weights, prompt engineering leverages clever input formatting to steer the model's behavior.

This is especially important because LLMs do not learn or change weights during inference. They only generate based on the prompt they are given. Prompt engineering gives users the ability to shape the model's behavior, style, tone, and output without retraining or fine-tuning.

Prompt engineering often involves:

- Instruction phrasing (e.g., “Explain like I’m five...”, “List three reasons...”)
- Inserting examples (few-shot learning)
- Using role-play (e.g., “You are a legal assistant...”)
- Formatting constraints (e.g., JSON output)

If no examples are given, the model does not invent any behind the scenes. That’s why prompt engineering matters: because the model depends entirely on how well the input is structured to produce desirable output.

Who or What Is the Assistant?

In the ChatGPT interface, the **assistant** refers to the language model's persona that generates responses. The model (e.g., GPT-4) is the underlying engine, and the assistant is the role it plays — defined by the system prompt.

```
You are ChatGPT, a helpful and knowledgeable assistant.
```

This defines how the model should behave when generating responses.

What Happens When You Ask ChatGPT a Question?

When you enter a question into ChatGPT, the following steps occur:

1. **System Prompt:** A hidden instruction like “You are ChatGPT, a helpful assistant” is prepended.
2. **User Input:** Your exact text is appended after the system prompt.
3. **Conversation History:** Prior user and assistant messages (within token limits) are included.
4. **Combined Prompt:** The full text is sent to the model to generate the next tokens autoregressively.

Does ChatGPT Modify or Preprocess the Input?

No. There is no explicit normalization (like punctuation removal or typo correction) before the model sees your input. The model handles informal input, typos, and unstructured language directly based on its training.

Is ChatGPT Doing Few-Shot Learning?

Yes — using a method called **in-context learning**. When you include examples in the prompt, the model infers the pattern and continues accordingly.

How Are Few-Shot Examples Constructed?

Few-shot examples are typically written manually by the user:

```
Input:  What is 2 + 2?  
Output: 4
```

```
Input:  What is 3 + 5?  
Output:
```

If you don't include examples, ChatGPT does not add them for you.

Input and Output Limits

- **Input (context window):** 4,096 tokens (GPT-3.5), 8K or 32K (GPT-4 variants)
- **Output:** Typically a few hundred to a few thousand tokens

Understanding the Full Pipeline

To fully understand the LLM input-output process:

- **Tokenization:** Input text is split into tokens (e.g., via BPE).
- **Embedding Layer:** Tokens are mapped to high-dimensional vectors.
- **Positional Encoding:** Adds position info to token vectors.
- **Transformer Blocks:** Layers of self-attention + feed-forward networks.
- **Output Layer:** Projects final hidden states to vocab size, applies softmax.
- **Decoding:** Generates next tokens using sampling, greedy, or beam search.

Common Prompt Engineering Techniques

Zero-Shot Prompting

Ask the model to perform a task without providing any examples.

Example:

Summarize the following paragraph: [paragraph here]

Few-Shot Prompting

Provide input-output examples in the prompt.

Example:

Translate English to Spanish:
Hello → Hola
Good morning → Buenos días
Thank you →

Chain-of-Thought Prompting

Encourage multi-step reasoning with prompts like “Let’s think step by step.”

Example:

Q: If there are 5 apples and you eat 2, how many are left?
A: Let’s think step by step.
First, there are 5 apples. You eat 2. So $5 - 2 = 3$.
Answer: 3

Role-Based Prompting

Assign a persona to the model.

Example:

You are a professional legal advisor. Explain this clause in plain English:

Instruction Prompting

Use direct, natural language instructions.

Example:

List three reasons why exercise is good for mental health.

Output Formatting Constraints

Ask the model to follow structured formats like JSON or tables.

Example:

Summarize the data in this JSON format.

Self-Consistency Prompting

Run multiple completions and choose the most consistent result (often used with chain-of-thought).

Use case: Improves accuracy and stability in tasks requiring reasoning.